

Query Analysis

Table 1: Display Courses Table

SQL Query

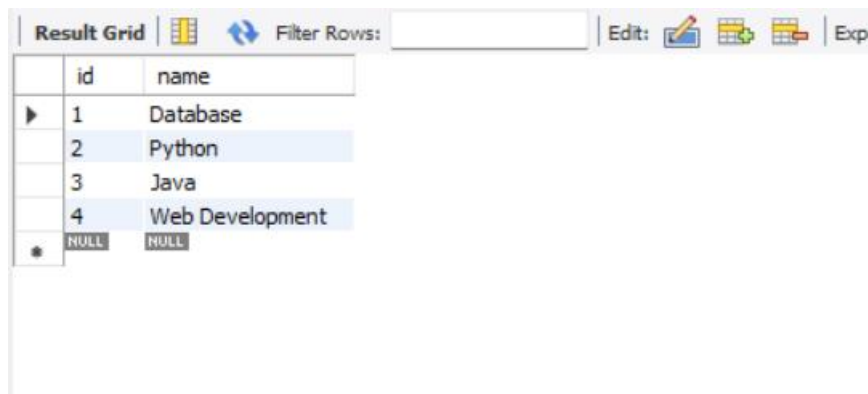
SELECT * FROM courses;

Result

Displays all courses available in the database along with their unique Course IDs.

Explanation

This query retrieves all records from the Courses table. It is used to verify that the course details have been inserted successfully into the database.



	id	name
▶	1	Database
	2	Python
	3	Java
	4	Web Development
*	NULL	NULL

Table 2: Display Enrollments Table

SQL Query

SELECT * FROM enrollments;

Result

Displays all student enrollment records, including the Student ID, Course ID, and the grade obtained in each course.

Explanation

This query retrieves every record from the Enrollments table. It verifies that students have been correctly linked to their respective courses and that their grades have been stored successfully.

	student_id	course_id	grade
▶	1	1	88
	2	2	75
	3	3	91
	4	4	82
	5	1	39
	6	2	85
	7	3	78
	8	4	95
	9	1	35
	10	2	90

Query 1: List All Students Enrolled in Each Course

SQL Query

SELECT

s.StudentName,

c.name AS CourseName,

e.grade

FROM enrollments e

JOIN students s

ON e.student_id = s.StudentsID

JOIN courses c

ON e.course_id = c.id

ORDER BY c.name, s.StudentName;

Result

Displays the name of each student, the course they are enrolled in, and their corresponding grade.

Explanation

This query uses INNER JOIN to combine the Students, Courses, and Enrollments tables. It retrieves student names, course names, and grades and sorts the results alphabetically by course and student name.

	Student_Name	Course_Name
►	Vishva	Database
	Neha	Database
	Pooja	Database
	Meet	Python
	Rahul	Python
	Rohan	Python
	Saxi	Java
	Arjun	Java
	Yashvi	Web Development
	Kavya	Web Development

Query 2: Average Grade Per Course

SQL Query

SELECT

c.name AS CourseName,

ROUND(AVG(e.grade),2) AS Average_Grade

FROM enrollments e

JOIN courses c

ON e.course_id = c.id

GROUP BY c.name

ORDER BY Average_Grade DESC;

Result

Displays the average grade obtained by students in each course.

Explanation

This query calculates the average grade for every course using the AVG() aggregate function. The GROUP BY clause groups the records according to each course, and ROUND() formats the average grade to two decimal places.

	Course_Name	Average_Grade
▶	Database	54.0000
	Python	83.3333
	Java	84.5000
	Web Development	88.5000

Query 3: Top 3 Students Overall

SQL Query

SELECT

s.StudentName,

ROUND(AVG(e.grade),2) AS Average_Grade

FROM enrollments e

JOIN students s

ON e.student_id = s.StudentsID

GROUP BY s.StudentName

ORDER BY Average_Grade DESC

LIMIT 3;

Result

Displays the top three students based on their average grades across all enrolled courses.

Explanation

This query calculates each student's average grade using the `AVG()` function. The records are sorted in descending order, and the `LIMIT` clause returns only the top three students.

	Name	Average_Grade
▶	Kavya	95.0000
	Saxi	91.0000
	Rohan	90.0000

Query 4: Count Students Who Failed

SQL Query

SELECT

COUNT(DISTINCT student_id) AS Failed_Students

FROM enrollments

WHERE grade < 40;

Result

Displays the total number of students who scored below 40 in at least one course.

Explanation

This query uses the `COUNT(DISTINCT)` function to count unique students who failed at least one course. The `WHERE` clause filters only those records where the grade is less than 40.

	Failed_Students
▶	2